

Big Data Analysis Exam Project

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Data Science study programme

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1. Objective

The objective of this project was to identify a vessel collision within the Baltic Sea of a specific time period. The objective has been achieved and a collision between two ships has been found and identified using big data analysis techniques. The analysis was conducted on the following time period: December 2021. The data analysis pipeline was restricted to vessels operating within a 50 nautical mile radius of latitude 55.225000 and longitude 14.245000.

The final output includes the two vessels, collision timestamp, collision coordinates, closest distance, and an HTML trajectory map covering 10 minutes before and 10 minutes after the collision time.

Dataset link: <http://aisdata.ais.dk/2021/aisdk-2021-12.zip>

2. Dataset

The dataset is the Danish AIS archive. This dataset contains one CSV file per day for December 2021 (31 days in total).

The following features/columns needed for collision detection and reporting were used for the analysis:

- timestamp,
- mmsi,
- latitude,
- longitude,
- navigational_status,
- sog,
- cog,
- heading,
- name,

- ship_type,
- width,
- length,
- draught

All other columns within the dataset were removed during the initial data processing steps to speed up the pipeline and reduce the resource usage.

3. Processing Methodology

The solution uses PySpark for the data processing pipeline. Intermediate datasets are written as Parquet folders, since PySpark is optimized to use these datasets.

The archive is processed one daily CSV at a time:

1. Extracts one daily CSV into `data/temp`.
2. Reads it.
3. Applies basic validity filtering and broad geographic filtering.
4. Appends the cleaned daily result to Parquet.
5. Deletes the extracted CSV.
6. Continues the process again with another day until all of the data in the collected data has been processed.

3.1 Initial Data Cleaning

To ensure that no invalid data is used for the subsequent analysis and that only the required data is saved, an initial data cleaning step was conducted. The first filtering step keeps only records satisfying the following conditions:

- Invalid MMSI/dirty data trap:
 - MMSI is exactly 9 digits.
 - MMSI is not 123456789.
 - MMSI is not one repeated digit, such as 111111111.
 - MMSI MID prefix is valid.
- Faulty data:
 - Timestamp can be parsed successfully.
 - Latitude is between -90 and 90.
 - Longitude is between -180 and 180.

3.2 Geographic Filtering

The assignment area is a 50 nautical mile radius around latitude 55.225000 and longitude 14.245000. The data processing pipeline is split to handle this requirement in two steps.

First, during daily processing, a broad 75 nautical mile pre-filter is applied. This helped with reducing the initial dataset size to a comparatively smaller size to speed up calculations.

After the initial GPS cleaning, the exact 50 nautical mile area filter is applied. This is a more in-depth method, since the data required for processing is much smaller.

Both filters use a fast bounding box first, and then finishes the calculations with an exact Haversine distance calculation.

3.3 GPS Noise Removal

AIS data can contain sudden GPS jumps that look like impossible vessel movement. These points can create false collision candidates if not removed.

GPS anomaly removal is performed for every single vessel:

1. Repartition records by MMSI.
2. Order each vessel track by timestamp.
3. Use Spark window functions to compare each point with the previous and next points.
4. Calculate the implied speed between neighboring positions using Haversine distance and timestamp difference.

An internal point is removed as an isolated GPS spike when both neighboring speeds exceed 60 knots.]

The result is then saved to another temporary folder to enforce segmentation and make the experiments easier. The later pipeline step uses this folder now.

3.4 Candidate Generation

A naive join between all vessel positions would be computationally expensive, so the pipeline uses time and spatial bucketing.

Each AIS point is assigned:

- a 1 minute time bucket;
- a spatial grid cell based on latitude and longitude.

Candidate pairs are found by joining only records in the same or neighboring time and spatial cells. The pair order is normalized so that each MMSI pair appears once. Exact Haversine distance is then calculated, and pairs are kept when the distance is at most 0.05 nautical miles.

3.5 Collision Validation

The candidate generation step intentionally keeps close encounters broadly. The next step is to apply more exact filters to find vessels that are much more likely to have collided.

For each candidate at time T , the pipeline inspects each vessel from $T - 10$ minutes to T , and from T to $T + 10$ minutes using the cleaned dataset. The dataset is used so that boundary filtering does not create fake gaps in the 20 minute trajectory window.

A candidate is retained only if all of the following rules are satisfied:

Table 1: Collision Validation Rules

Rule	Threshold
Closest distance	≤ 5 meters
Observed combined speed drop	≥ 0.2
Pre event AIS points for each vessel	≥ 20
Pre event movement for each vessel	≥ 1000 meters
Approach distance	≥ 1000 meters
Excluded ship type	neither vessel is Pilot

Closes distance is the default requirement - only vessels that were extremely close to each other were flagged. Additionally, if the general speed of both of the vessels have dropped in 20 %, then it was also flagged.

Additionally, after an initial analysis it was found that at least two ports are within the 50 nautical miles of the starting location.

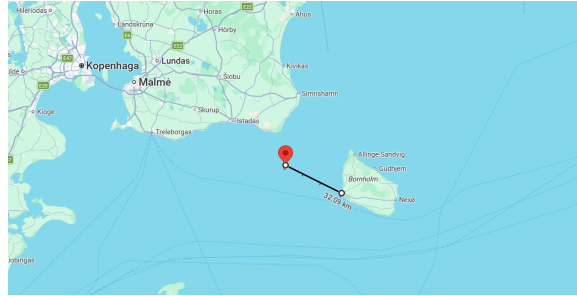


Figure 1: Starting point vs closest port distance calculation

Thus it was important to also validate if the vessels were actually moving before the event. Thus three more rules were added to disregard such events:

- Vessels that were not transmitting more than 20 points before the collision.
- Vessels that were not moving more than 1km before the collision.
- Vessels that were closer than 1km to each before the collision.

Lastly, the pilot vessels were excluded due to them being the outliers of this detection algorithm. Since there were ports within the scanned sea area, there were pilot vessels.

Since their purpose is to guide the cargo ships by getting extremely close to them, they were likely to trigger this detection flow without actually having a crash.

Since the pilot vessels were designed to handle collisions and due to the previously mentioned reasons, it was determined to exclude this vessel type in this analysis.

4. Results

The final detected collision is:

Table 2: Detected Collision Details

Field	Value
Vessel 1 MMSI	219021240
Vessel 1 name	KARIN HOEJ
Vessel 1 ship type	Other
Vessel 2 MMSI	232018267
Vessel 2 name	MV SCOT CARRIER
Vessel 2 ship type	Cargo
Collision time	2021-12-13 02:27:43
Collision latitude	55.2230795
Collision longitude	14.2437065
Minimum distance	4.08 meters

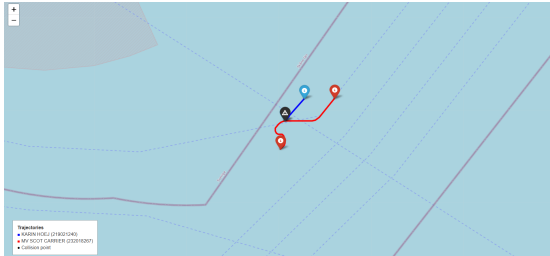


Figure 2: Trajectory of the two identified vessels

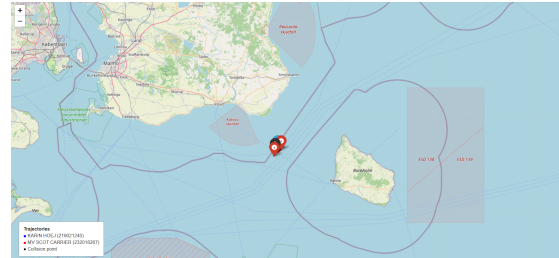


Figure 3: Trajectory of the two identified vessels (expanded)

The pre-event movement supports the collision interpretation. KARIN HOEJ had 55 AIS points before the event, a median pre-event SOG of 6.0 knots, and moved approximately 1788 meters. MV SCOT CARRIER had 96 AIS points before the event, a median pre-event SOG of 11.8 knots, and moved approximately 3278 meters.

The vessels were approximately 1830 meters apart at the start of the 10 minute pre-event window and reduced that separation to approximately 4 meters. This gives an approach distance of approximately 1826 meters.

After the close encounter, MV SCOT CARRIER continued transmitting and had an observed median SOG of 8.8 knots. KARIN HOEJ had no after-event AIS points in the 10

minute window. This dropout is reported in the data through the after-event point count, but it is not used as a filtering rule.

A rudimentary internet analysis has uncovered the instance report of the collision confirming that this is indeed the collision instance:

- Wikipedia: https://en.wikipedia.org/wiki/2021_Baltic_Sea_incident
- Marine Accident Investigation Branch incident report: <https://assets.publishing.service.gov.uk/media/64f9bbec9ee0f2000fb7c054/2023-5-ScotCarrier-KarinHoej-ReportAndAnalysis.pdf>

5. Computational Decisions

Several design choices reduce computational cost:

- Processing one daily CSV at a time avoids storing all uncompressed raw files.
- Column pruning reduces memory and storage costs.
- A broad 75 nautical mile pre-filter reduces data before vessel-track sorting.
- Parquet is used for temporary outputs, allowing efficient data handling.
- GPS jump removal uses partitioned vessel windows instead of loops.
- Candidate detection uses time buckets and spatial grid cells.
- Exact Haversine distance is calculated only after cheaper time and grid filters.

6. Conclusion

The pipeline has correctly identified KARIN HOEJ and MV SCOT CARRIER as the collision pair. The vessels came within approximately 4.08 meters at 2021-12-13 02:27:43 near latitude 55.2230795 and longitude 14.2437065.